

GED Math Unit: Number Sense & Problem Solving

Prepared for Instruction

November 6, 2025

Objectives

- Identify and classify number types: whole numbers, integers, rationals, irrationals.
- Convert between fractions, decimals, and percentages.
- Use place value to round numbers appropriately.

Key Ideas

- Place value: ones (10^{-1}), hundredths (10^{-2}), thousandths (10^{-3}), ...
- Conversion: percent $\rightarrow$ decimal by dividing by 100; decimal $\rightarrow$ fraction by reading place value and simplifying.
- Number types: integers ($\dots, -2, -1, 0, 1, 2, \dots$), rational ($\frac{a}{b}, b \neq 0$), irrational (non-repeating, non-terminating decimals).

Worked Example 1

Convert 0.75 to a fraction.

$$0.75 = \frac{75}{100} = \frac{3}{4}.$$

Answer: $\frac{3}{4}$.

Worked Example 2

Simplify $\frac{18}{24}$.

$$\begin{aligned}\gcd(18, 24) &= 6, \\ \frac{18}{24} &= \frac{18 \div 6}{24 \div 6} = \frac{3}{4}.\end{aligned}$$

Answer: $\frac{3}{4}$.

Worked Example 3

Round 0.03456 to the nearest thousandth.

Thousandth place = 3rd decimal digit $\rightarrow$ 0.034 (look at next digit 5, round up) $\Rightarrow$ 0.035.

Answer: 0.035.

Worked Example 4

Classify -7 , 3.14 , $\frac{1}{2}$.

- -7 : integer (also rational).
- 3.14 : terminating decimal $\Rightarrow$ rational.
- $\frac{1}{2}$: rational; as decimal 0.5 .

Worked Example 5

Convert 125% **to a decimal and fraction.**

$$125\% = \frac{125}{100} = 1.25 = \frac{125}{100} = \frac{5}{4}.$$

Answer: decimal 1.25, fraction $\frac{5}{4}$.

Objectives

- Estimate sums, differences, products, and quotients.
- Use compatible numbers and front-end estimation.
- Judge reasonableness of answers.

Worked Example 1

Estimate $234 + 567$.

$$234 \approx 200, \quad 567 \approx 600 \Rightarrow 200 + 600 = 800.$$

Reasonable exact sum = 801, so estimate 800 is close.

Worked Example 2

Estimate 82×49 using compatible numbers.

$$82 \approx 80, \quad 49 \approx 50 \Rightarrow 80 \times 50 = 4000.$$

Reasonableness: exact $82 \times 49 = 4018$.

Worked Example 3

Approximate $1278 \div 6$.

$$1278 \approx 1200, \quad 1200 \div 6 = 200.$$

Exact = 213, so estimate 200 is reasonable.

Worked Example 4

Is $49 \times 21 \approx 1000$ **reasonable**?

$50 \times 20 = 1000$ (close), exact = 1029. Yes, reasonable.

Worked Example 5

Round 23.746 to nearest tenth and whole.

Tenth: look at hundredths (4) $\Rightarrow$ 23.7; Whole: look at tenths (7) $\Rightarrow$ 24.

Objectives

- Write and simplify ratios.
- Solve proportions by cross-multiplication.
- Compute percent, percent change, and discounts.

Worked Example 1

Write and simplify the ratio of 6 apples to 9 oranges.

$$6 : 9 = \frac{6}{9} = \frac{2}{3} \Rightarrow 2 : 3.$$

Worked Example 2

Solve $\frac{3}{4} = \frac{x}{16}$.

$$3 \cdot 16 = 4x \Rightarrow 48 = 4x \Rightarrow x = 12.$$

Worked Example 3

Find 25% **of** 120.

$$25\% = \frac{1}{4} \Rightarrow \frac{1}{4} \cdot 120 = 30.$$

Worked Example 4

Price increases from \$40 to \$50. Find percent increase.

$$\text{Change} = 50 - 40 = 10, \quad \frac{10}{40} = 0.25 = 25\%.$$

Worked Example 5

\$30 shirt on sale for 20% off. Find sale price.

$$\text{Discount} = 0.20 \times 30 = 6, \quad \text{Sale price} = 30 - 6 = \$24.$$

Objectives

- Apply a four-step process: Understand, Plan, Solve, Check.
- Translate word problems into equations.
- Evaluate the reasonableness of solutions.

Worked Example 1

“Twice a number is 10 more than 8.” Find the number.

$$2x = 8 + 10 = 18 \Rightarrow x = 9.$$

Check: $2 \cdot 9 = 18$ matches.

Worked Example 2

Recipe uses 3 cups flour for 12 cookies. How much for 20?

$$\text{Rate} = \frac{3}{12} = \frac{1}{4} \text{ cup per cookie} \Rightarrow 20 \times \frac{1}{4} = 5 \text{ cups.}$$

Worked Example 3

5 pencils cost \$2.25. Cost for 12 pencils?

$$\text{Unit price} = \frac{2.25}{5} = \$0.45, \quad 12 \times 0.45 = \$5.40.$$

Worked Example 4

Earn \$15/hr for 35 hours. How much after 15% tax?

$$\text{Gross} = 15 \times 35 = \$525, \quad \text{Tax} = 0.15 \times 525 = \$78.75,$$

$$\text{Net} = 525 - 78.75 = \$446.25.$$

Worked Example 5

Car travels 240 miles in 4 hours. Find speed. Then test 250 miles in 4.5 hrs.

$$\text{Speed} = \frac{240}{4} = 60 \text{ mph.}$$

At 60 mph for 4.5 h $\Rightarrow 60 \times 4.5 = 270$ mi. 250 mi is reasonable but a bit less.

Objectives

- Model mixture, distance-rate-time, work, and money problems.
- Interpret real-world scenarios and select operations.

Worked Example 1

Distance: A car travels 60 mph for 3 hours. How far?

$$d = rt = 60 \times 3 = 180 \text{ miles.}$$

Worked Example 2

Tank holds 50 L and is 40% full. How many liters?

$$0.40 \times 50 = 20 \text{ L.}$$

Worked Example 3

Two workers finish a job in 6 h. One alone takes 10 h. How long for the other alone?

Let rates be $A = \frac{1}{10}$ job/h and $B = \frac{1}{x}$ job/h.

$$A + B = \frac{1}{6} \Rightarrow \frac{1}{10} + \frac{1}{x} = \frac{1}{6} \Rightarrow \frac{1}{x} = \frac{1}{6} - \frac{1}{10} = \frac{5-3}{30} = \frac{2}{30} = \frac{1}{15}.$$

Thus $x = 15$ hours.

Worked Example 4

8% sales tax on a \$45 item. What is the total?

$$\text{Tax} = 0.08 \times 45 = \$3.60, \quad \text{Total} = 45 + 3.60 = \$48.60.$$

Worked Example 5

Reading a bar graph of monthly spending. Identify highest category.

- Strategy: Compare bar heights (or values) and select the maximum.
- Always read axes labels and units before comparing.

(Insert class-specific data graphic if available.)

Practice Prompts (Optional Add-on)

- 1 Convert 0.375 to a fraction in simplest form.
- 2 Estimate 389×21 using compatible numbers.
- 3 Solve $\frac{7}{8} = \frac{y}{24}$.
- 4 A jacket listed at \$64 has 30% off and 8% tax. Find the total price.
- 5 A train travels 45 mph. How long to go 157.5 miles?

Exit Ticket

- One concept I mastered today is _____.
- One question I still have is _____.
- My strategy for solving word problems is _____.